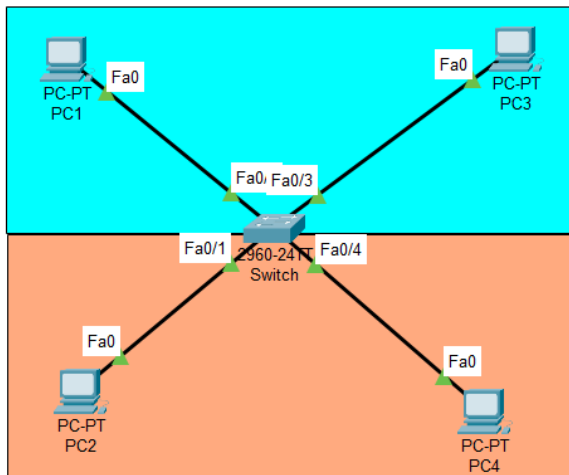


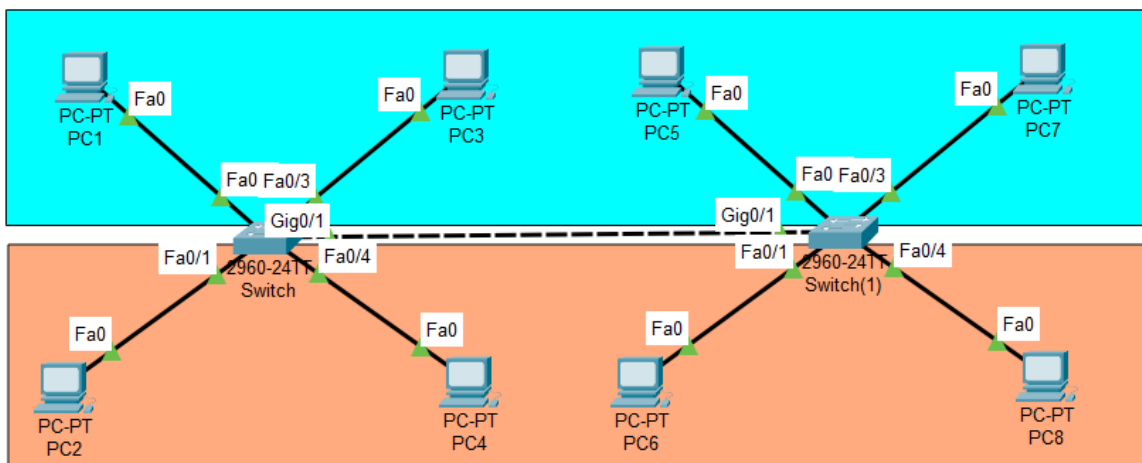
# Practice 3 – VLAN Technology

## Topology

### Part 1: Access port



### Part 2: Trunk port



## Objectives

Part 1: Access Port Configuration

Part 2: Trunk Port Configuration

## Background

VLAN (*Virtual Local Area Network*) is helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access and trunk ports to specific VLANs.

*Access ports* are for end devices, while *Trunk ports* are for inter-switch or inter-network communication.

## Part 1: Access port's topology

### Configuring VLANs

#### Access port

**Step 1:** Access the CLI of the switch and enter the **configuration mode**:

```
Switch> enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

**Step 2:** Create VLANs.

- The topology is divided into two VLANs. By default, all switch ports belong to VLAN 1, so we need to create a new one.

- To create new VLAN, use the **vlan [num]** command:

```
Switch(config)# vlan 2
Switch(config-vlan)#
```

- Give to vlan 2 name, by using **name [name]** command:

```
Switch(config-vlan)# name Bob
Switch(config-vlan)# exit
```

- Repeat commands for vlan 3.

```
Switch(config)# vlan 3
Switch(config-vlan)# name Alice
Switch(config-vlan)# exit
```

### Step 3: Configuring interfaces

In VLAN 2, according to the topology, there are two computers: **PC1** and **PC3**. They are connected to the switch through **Fa0/2** and **Fa0/3**.

- Access the configuration for interface **Fa0/2**: `interface fastEthernet 0/2`  
`Switch(config)#interface fastEthernet 0/2`
- Set the Port to **Access mode**: `switchport mode access`  
`Switch(config-if)#switchport mode access`
- Assign the port to **vlan 2**: `switchport access vlan 2`  
`Switch(config-if)#switchport access vlan 2`  
`Switch(config-if)#exit`
- Repeat the same configuration for Fa0/3:  
`Switch(config)#interface fastEthernet 0/3`  
`Switch(config-if)#switchport mode access`  
`Switch(config-if)#switchport access vlan 2`  
`Switch(config-if)#exit`  
`Switch(config)#end`

In VLAN 3, according to the topology, there are two computers: **PC2** and **PC4**. They are connected to the switch through **Fa0/1** and **Fa0/4**. Repeat steps above:

```
Switch(config)#vlan 3
Switch(config-vlan)#name Alice
Switch(config-vlan)#exit
```

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
```

```
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#end
```

To check VLAN configuration, use the command: **show vlan brief**

```
Switch#show vlan brief
```

| VLAN | Name    | Status | Ports                                                                                                                                                                     |
|------|---------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default | active | Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2 |
| 2    | Bob     | active | Fa0/2, Fa0/3                                                                                                                                                              |
| 3    | Alice   | active | Fa0/1, Fa0/4                                                                                                                                                              |

-----More-----

**Step 4:** Save the configuration to memory: **write memory** or shortly **wr mem**

```
Switch#wr mem
```

```
Building configuration...
```

```
[OK]
```

```
Switch#
```

**Step 5:** Configure IP addresses for all PCs.

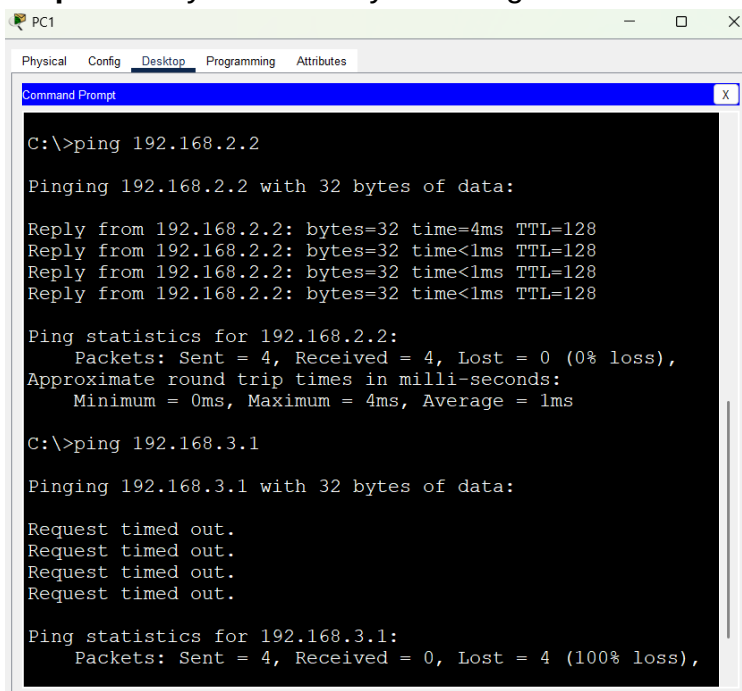
IP-address of PC1: 192.168.2.1 255.255.255.0

IP-address of PC2: 192.168.3.1 255.255.255.0

IP-address of PC3: 192.168.2.2 255.255.255.0

IP-address of PC4: 192.168.3.2 255.255.255.0

**Step 6:** Verify Connectivity with Ping.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

## Part 2: Trunk port's topology

### Trunk topology

Duplicate the topology from Part 1 and connect two switches using a **Copper Cross-Over Cable** to the *GigabitEthernet0/1* ports. Rename the PCs according to the new setup, and assign new IP addresses for PCs as follows:

IP-address of PC5: 192.168.2.3 255.255.255.0

IP-address of PC6: 192.168.3.3 255.255.255.0

IP-address of PC7: 192.168.2.4 255.255.255.0

IP-address of PC8: 192.168.3.4 255.255.255.0

If the configuration on the right switch didn't save, repeat all steps [above](#) (Step1-4).

Left switch:

**Step 1: Configure Trunk Port on the GigabitEthernet0/1 Interface.**

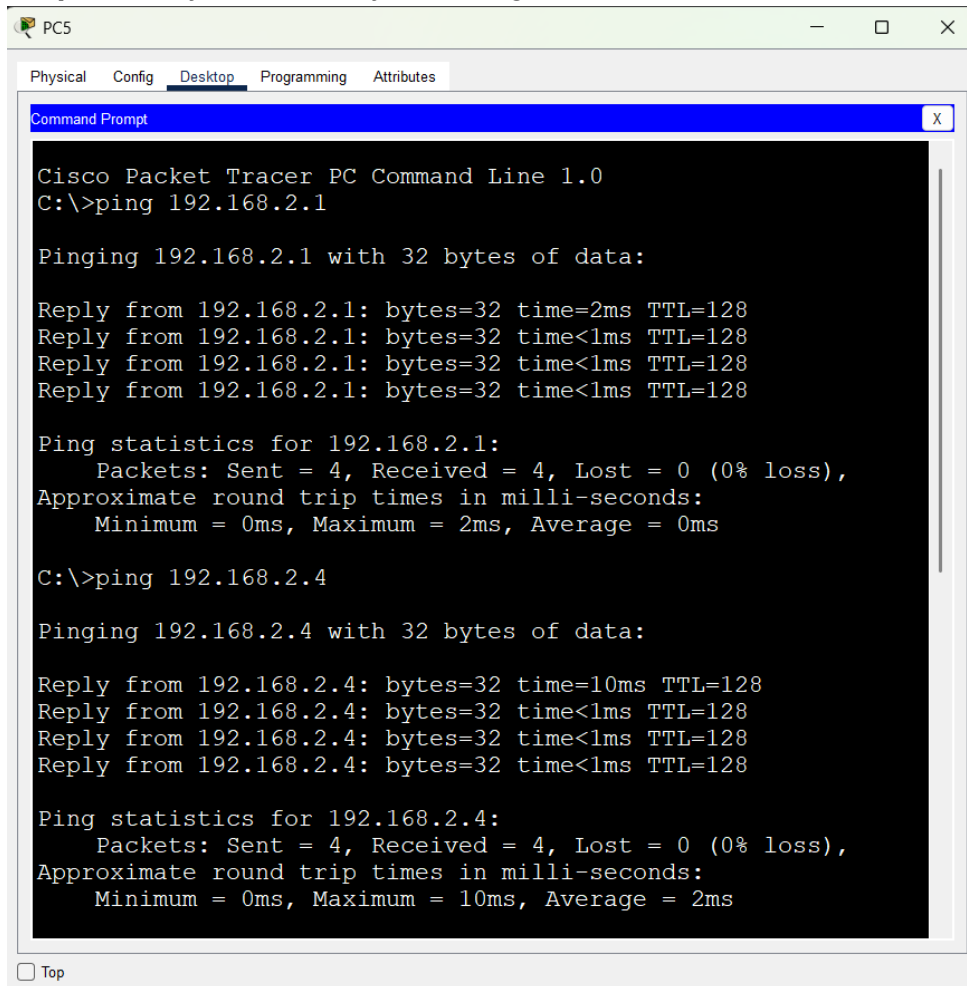
- Select the Interface to Configure: **interface gigabitEthernet0/1**  
Switch(config)#**interface gigabitEthernet 0/1**
- Set the interface to **Trunk Mode**:  
Switch(config-if)#**switchport mode trunk**  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up  
Switch(config-if)#
- Define which VLANs can pass through this trunk port:  
Switch(config-if)#**switchport trunk allowed vlan 2,3**  
Switch(config-if)#**exit**  
Switch(config)#**end**
- Save configurations  
Switch#**wr mem**  
Building configuration...  
[OK]  
Switch#

**!** Repeat commands to the **right switch**.

**Optional Step:** To remove a VLAN from the trunk:

```
Switch(config)# interface gigabitEthernet 0/1  
Switch(config-if)# switchport trunk allowed vlan remove  
20
```

## Step 2: Verify Connectivity with Ping:



PC5

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Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=2ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

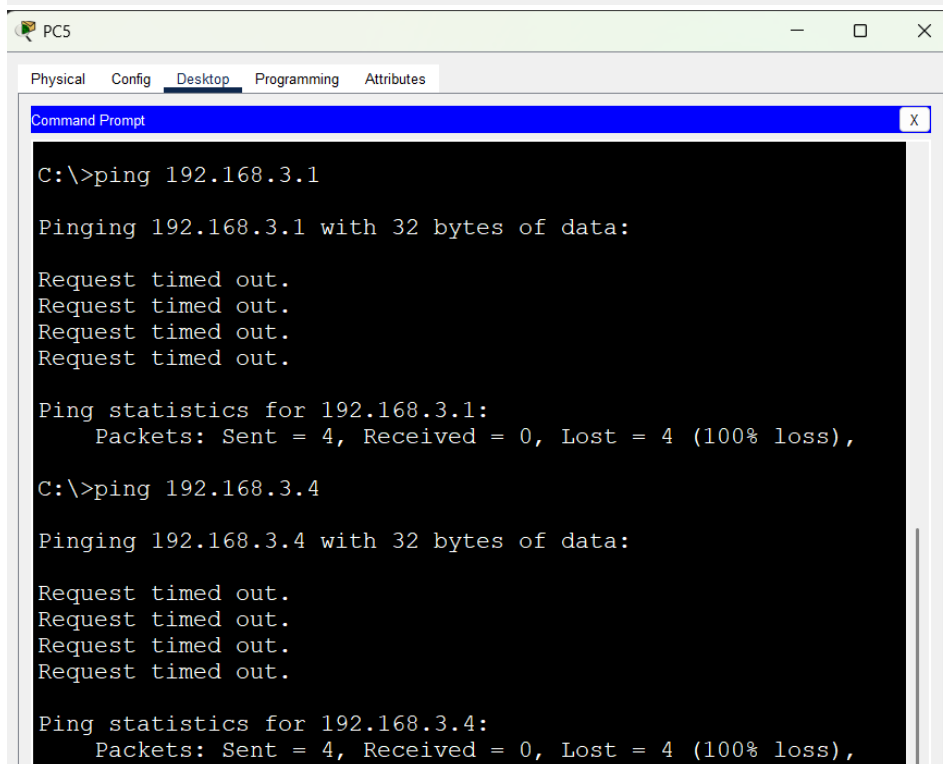
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=10ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

☐ Top



PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.3.4

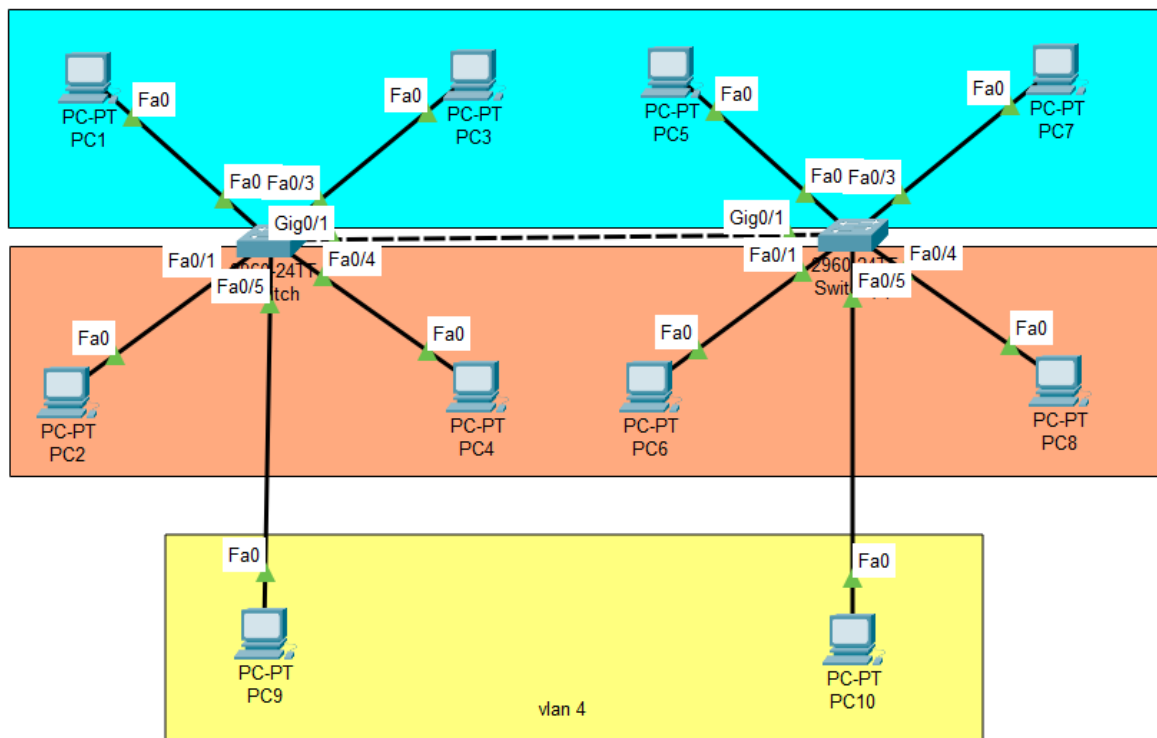
Pinging 192.168.3.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

# Practical Task

Connect another computer to each switch and configure vlan 4.



**Hint:** Don't forget to configure the IP address for the PC: 192.168.4.1 255.255.255.0 and 192.168.4.2 255.255.255.0